

Exercises 32.1-1 to 32.1-4

Questions and Solutions One by One

Naive String-Matching Algorithm

1 Exercise 32.1-1

Question 32.1-1

Show the comparisons the naive string matcher makes for the pattern

$$P = 0001$$

in the text

$$T = 000010001010001.$$

Solution

We have

$$P = 0001, \quad m = 4$$

and

$$T = 000010001010001, \quad n = 15.$$

So the possible shifts are

$$s = 0, 1, 2, \dots, 11.$$

At each shift, the naive matcher compares from left to right and stops at the first mismatch.

Shift s	Text window	Comparisons made	Count
0	0000	$0 = 0, 0 = 0, 0 = 0, 1 \neq 0$	4
1	0001	$0 = 0, 0 = 0, 0 = 0, 1 = 1$	4
2	0010	$0 = 0, 0 = 0, 0 \neq 1$	3
3	0100	$0 = 0, 0 \neq 1$	2
4	1000	$0 \neq 1$	1
5	0001	$0 = 0, 0 = 0, 0 = 0, 1 = 1$	4
6	0010	$0 = 0, 0 = 0, 0 \neq 1$	3
7	0101	$0 = 0, 0 \neq 1$	2
8	1010	$0 \neq 1$	1
9	0100	$0 = 0, 0 \neq 1$	2
10	1000	$0 \neq 1$	1
11	0001	$0 = 0, 0 = 0, 0 = 0, 1 = 1$	4

Therefore, the valid shifts are

$$s = 1, 5, 11.$$

Total comparisons:

$$4 + 4 + 3 + 2 + 1 + 4 + 3 + 2 + 1 + 2 + 1 + 4 = \boxed{31}.$$

2 Exercise 32.1-2

Question 32.1-2

Suppose that all characters in the pattern P are different. Show how to accelerate NAIVE-STRING-MATCHER to run in time $\mathcal{O}(n)$ on an n -character text T .

Solution

Assume all characters of the pattern are distinct.

Suppose at shift s , the first q characters match:

$$P[1] = T[s + 1], P[2] = T[s + 2], \dots, P[q] = T[s + q],$$

but the next comparison fails:

$$P[q + 1] \neq T[s + q + 1].$$

Because all characters in P are different, the matched text characters

$$T[s + 1], T[s + 2], \dots, T[s + q]$$

correspond to

$$P[1], P[2], \dots, P[q],$$

which are all distinct.

Now any shift

$$s + 1, s + 2, \dots, s + q - 1$$

cannot be valid. The reason is that at any such shift, the first character $P[1]$ would align with one of the already matched characters $P[2], P[3], \dots, P[q]$, and none of them equals $P[1]$.

So after matching q characters and then failing, we can shift by q directly.

Improved Rule

- If no character matched, shift by 1.
- If $q > 0$ characters matched before mismatch, shift by q .

Algorithm

DISTINCT-NAIVE-MATCHER(T, P)

```
1  n = |T|, m = |P|
2  s = 0
3  while s <= n - m
4      q = 0
5      while q < m and P[q+1] == T[s+q+1]
6          q = q + 1
7      if q == m
8          print "Pattern occurs with shift", s
9          s = s + m
10     else if q == 0
11         s = s + 1
12     else
13         s = s + q
```

Running Time

Each time the algorithm makes successful comparisons, it later skips forward by exactly that many positions. So the total amount of work is linear in the text length.

Hence the running time is

$$\mathcal{O}(n).$$

3 Exercise 32.1-3

Question 32.1-3

Suppose that pattern P and text T are randomly chosen strings of length m and n , respectively, from the d -ary alphabet

$$\Sigma_d = \{0, 1, \dots, d-1\}, \quad d \geq 2.$$

Show that the expected number of character-to-character comparisons made by the implicit loop in line 4 of the naive algorithm is

$$(n - m + 1) \frac{1 - d^{-m}}{1 - d^{-1}} \leq 2(n - m + 1).$$

Solution

There are

$$n - m + 1$$

possible shifts.

Let C be the number of comparisons made at one fixed shift.

The i -th comparison happens only if the first $i-1$ characters match. Since the strings are random and independent,

$$\mathbb{P}(\text{first } i-1 \text{ characters match}) = d^{-(i-1)}.$$

So

$$\mathbb{P}(C \geq i) = d^{-(i-1)}, \quad i = 1, 2, \dots, m.$$

Using

$$\mathbb{E}[C] = \sum_{i=1}^m \mathbb{P}(C \geq i),$$

we get

$$\mathbb{E}[C] = \sum_{i=1}^m d^{-(i-1)} = \sum_{k=0}^{m-1} d^{-k}.$$

This is a geometric series:

$$\mathbb{E}[C] = \frac{1 - d^{-m}}{1 - d^{-1}}.$$

Since there are $n - m + 1$ shifts, the expected total number of comparisons is

$$(n - m + 1) \frac{1 - d^{-m}}{1 - d^{-1}}.$$

Now show the upper bound. Since $d \geq 2$,

$$d^{-1} \leq \frac{1}{2}.$$

Therefore,

$$\frac{1}{1-d^{-1}} \leq \frac{1}{1-\frac{1}{2}} = 2.$$

Also,

$$1-d^{-m} \leq 1.$$

Hence,

$$\frac{1-d^{-m}}{1-d^{-1}} \leq 2.$$

Multiplying by $(n-m+1)$:

$$(n-m+1) \frac{1-d^{-m}}{1-d^{-1}} \leq 2(n-m+1).$$

Thus the naive string matcher is quite efficient on average for random strings.

4 Exercise 32.1-4

Question 32.1-4

Suppose we allow the pattern P to contain occurrences of a gap character

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that can match an arbitrary string of characters, even a string of length 0.

Give a polynomial-time algorithm to determine whether such a pattern P occurs in a given text T , and analyze the running time.

Solution

Let the gap symbol \ match any substring, including the empty substring.

Step 1: Split the pattern into blocks

Split the pattern at each gap symbol:

$$P = X_1 \setminus X_2 \setminus \cdots \setminus X_k,$$

where each X_i contains no gap symbol.

Example:

ab\ba\c

becomes

$$X_1 = \text{ab}, \quad X_2 = \text{ba}, \quad X_3 = \text{c}.$$

If the pattern contains only gap symbols, then it always matches the text.

Step 2: Matching idea

The pattern matches the text if and only if the blocks

$$X_1, X_2, \dots, X_k$$

occur in the text in this order, with anything allowed between them.

So we just need to find:

$$X_1 \text{ then } X_2 \text{ then } \cdots \text{ then } X_k.$$

Step 3: Greedy algorithm

Search for the blocks from left to right, always taking the earliest possible occurrence.

```
GAP-MATCH(T, P)
1  Split P into nonempty blocks X1, X2, ..., Xk
2  if k == 0
3      return true
4  pos = 1
5  for i = 1 to k
6      Find the leftmost occurrence of Xi in T starting at or after pos
7      if no such occurrence exists
8          return false
9      pos = ending position of that occurrence + 1
10 return true
```

Why it works

Choosing the earliest occurrence of each block leaves the maximum remaining space for later blocks. So if any valid full match exists, this greedy strategy will also succeed.

Running Time

Let

$$|T| = n, \quad |P| = m.$$

If we search each block using any polynomial-time substring search method, then a simple upper bound is

$$\mathcal{O}(kn + m),$$

where k is the number of blocks.

Since $k \leq m$, this is polynomial. In particular,

$$\boxed{\mathcal{O}(mn)}$$

is a valid bound.